



Regular Article

Reassessing blockchain in global development: Tokenization of infrastructure, social impacts, and the axie infinity case in the Philippines

Paolo Joseph Layosa Lising 

Harvard University, Extension School, United States

A B S T R A C T

Developing countries struggle to fund infrastructure projects. Researchers suggest blockchain-based tokenization as a potential solution, yet concerns about social inequality remain underexplored. Existing studies on tokenization focus primarily on conceptual insights and limited case studies. Meanwhile, empirical evidence on its social implications in developing economies is lacking. This study addresses the gap by conducting a meta-analysis of secondary datasets assessing the Philippines' readiness for infrastructure tokenization and a case study of Axie Infinity to examine the social risks of token-based economic models. Findings indicate that the Philippines faces low digital competitiveness, financial literacy challenges, and gender disparities that limit equitable benefits from blockchain adoption. Tokenization risks reinforcing wealth concentration, as seen in the case of Axie Infinity, where lower-income participants have suffered financial losses while wealthier individuals have benefited from playing the game. Regulatory and security gaps in the blockchain ecosystem further amplify these risks. At a time when the Philippines' Securities and Exchange Commission (SEC) is actively developing rules for crypto-related services through its draft SEC Rules on Crypto-Assets Service Providers (CASP), a deeper understanding of tokenization's social implications is essential. As the government takes steps to regulate this space, this study provides timely insights into the potential social risks and challenges that must be addressed to ensure equitable adoption. One perspective likens this technology to the Internet in the 1990s, suggesting that it will have a lasting presence and continue to shape access to information, education, and economic opportunities, particularly in developing countries. Given blockchain's inevitable expansion, critically examining its risks is essential to ensure it does not deepen existing inequalities, particularly in emerging economies.

1. Introduction

Developing countries struggle to fund essential infrastructure projects, and researchers from the World Bank and the Asian Development Bank see an emerging technology that uses blockchain, also known as the tokenization of infrastructure, as a potential solution (The World Bank Group, 2023; Schletz et al., 2020). Tokenization involves fractionalizing and representing real-world assets using tokens, or digital certificates, through blockchain technology (Berntsen & Leippold, 2022), and infrastructure tokenization makes tokenization available specifically to represent physical infrastructure assets (Tian et al., 2020; Uzsoki, 2019). Tokenization, which allows retail investors to acquire fractionalized shares of infrastructure (Tian, 2020; 2022; Uzsoki, 2019), enables ordinary people to participate in funding large-scale projects previously exclusive to institutional investors and banks. However, researchers have pointed out the potential of blockchain-based initiatives to increase social inequalities without careful design (Lapointe & Fishbane, 2018; Tian et al., 2022). Existing research on these societal effects is limited, primarily comprising conceptual insights and limited case studies in developing countries. In addressing the gap in the literature, the Philippines, recognized for its pioneering role in adopting

blockchain technology in Asia (Ungson & Soorapanth, 2022), will serve as an ideal case study for exploring the potential benefits and social impacts of blockchain adoption, particularly in infrastructure tokenization. Despite progressive regulatory initiatives, the Philippines faces challenges such as low financial literacy, limited digital competitiveness, and persistent gender disparities, all of which hinder the potential of blockchain technologies, including tokenization.

To explore these challenges, this study employs a two-pronged approach: a meta-analysis of secondary datasets and a case study of Axie Infinity. First, the meta-analysis synthesizes data from reputable sources, including the IMD World Digital Competitiveness Ranking (IMD, 2024), the OECD/INFE 2023 International Survey of Adult Financial Literacy (OECD, 2023), and the Chainalysis (2024) global crypto adoption index. These datasets were selected for their relevance in assessing the Philippines' capacity for adopting blockchain-based tokenization. Metrics, including digital literacy, financial literacy, and crypto adoption rates, are analyzed to identify systemic challenges and opportunities that are unique to the Philippines. Second, the Axie Infinity case study serves as a microcosm for understanding the social implications of tokenization. By examining media reports covering user testimonials and economic data, this study explores how the game's

E-mail addresses: pal793@g.harvard.edu, paolojoseph@gmail.com.

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mechanism impacted the Philippines' social equity and economic stability at an individual level. The study particularly analyzes the collapse of Axie Infinity to uncover the potential risks and lessons that apply to broader tokenization efforts, particularly in infrastructure development. These findings are summarized in a structured table presenting risks, evidence, and prescriptive insights specific to this case. The dual-method approach comprehensively explores data and empirical evidence that is currently missing from the literature on blockchain's potential and its social impact.

To situate this research within a broader context, the study will first review existing literature on blockchain technology, synthesizing conceptual analyses and case studies to outline the potential benefits and challenges of tokenizing infrastructure. Building on this foundation, the study outlines the origins of risk in blockchain-based applications to clarify potential points of failure and guide future interventions. These insights will inform the analysis of the Philippine case to assess tokenization's social implications and highlight the risks in the context of this developing economy. Furthermore, this analysis is crucial, as [Uzsoki \(2019\)](#) argues, "this technology is here to stay" (p. 26), much like "the Internet in the 1990s" (p. 26), underscoring the importance of understanding tokenization's broader impact on global development.

1.1. Tokenization of infrastructure: conceptual foundations

Tokenization, a process that involves representing real-world assets digitally through blockchain-based tokens, is increasingly being explored in the context of infrastructure financing ([Berntsen & Leippold, 2022](#)). This process operates on the blockchain, a type of Distributed Ledger Technology (DLT) that records data in cryptographically linked blocks distributed across networks ([Anthony Jr, 2023](#); [Chung et al., 2023](#); [Tian et al., 2020, 2022a](#); [The World Bank Group, 2023](#)). While blockchain and DLT are often treated synonymously, authors such as [Jiang \(2022\)](#), [Lehmann \(2021\)](#), and [Patel and Ganne \(2020\)](#) emphasize definitional distinctions, reflecting the still-evolving conceptual landscape of the technology. The application of tokenization to physical infrastructure, referred to as infrastructure tokenization, represents a novel form of digitizing traditionally illiquid assets, making them more divisible and accessible ([Tian et al., 2020](#); [Uzsoki, 2019](#)). Tokens used in this context can be categorized into three general types: payment tokens, which serve transactional purposes; utility tokens, which grant access to specific services; and security tokens, which confer ownership rights ([Nelson, 2019](#); [OECD, 2019](#); [Tian, 2022a](#); [The World Bank Group, 2023](#)). Some authors, such as [Ferreira \(2020\)](#) and [Kim \(2020\)](#), use the term "currency tokens" in place of "payment tokens," underscoring the ongoing terminological variability. This diversity in both terminology and token classification highlights the conceptual complexity that accompanies efforts to apply tokenization in infrastructure markets.

The tokenization of infrastructure is situated within a broader academic discussion on the potential of blockchain to improve the financing of capital-intensive projects. Scholars argue that blockchain applications can reduce transaction costs and increase transparency in infrastructure financing ([Berntsen & Leippold, 2022](#); [Chung et al., 2023](#); [Neves & Prata, 2018](#); [OECD, 2019](#); [Schletz, 2020](#); [Tian, 2020, 2022a, 2022b](#); [Uzsoki, 2019](#); [The World Bank Group, 2023](#)). From a global development finance perspective, the [World Bank Group \(2023\)](#) acknowledges potential short-term costs associated with tokenization initiatives, even as it continues to explore blockchain applications. [Schletz \(2020\)](#) emphasizes that real-world implementation, rather than theory alone, is needed to evaluate the viability of infrastructure tokenization in practice.

1.2. Exploring tokenization's potential in infrastructure funding

Multiple researchers have discussed the benefits of the tokenization of infrastructure, offering insights from a theoretical perspective and limited real-life use cases ([Chung et al., 2023](#); [Heines & Jung, 2021](#); [Tian](#)

[et al., 2020, 2022a, 2022b](#); [Uzsoki, 2019](#); [Schletz et al., 2020](#); [The World Bank Group, 2023](#)). In particular, blockchain's immutable records are highlighted as a key feature that can improve transparency by ensuring tamper-proof documentation and tracking of transactions and project progress ([Tian et al., 2020, 2022a, 2022b](#); [Uzsoki, 2019](#)). Decentralization is another benefit of tokenization, as it allows people to trade the tokenized assets directly with each other without needing a middleman, reducing the risk of relying on others to manage transactions, as noted in the papers by [Uzsoki \(2019\)](#), [Heines and Jung \(2021\)](#), and [Chung et al. \(2023\)](#).

The literature on tokenization includes various examples of its applications. Real estate is the most frequently highlighted application of tokenization, mentioned by [Heines and Jung \(2021\)](#), [Schletz et al. \(2020\)](#), [Uzsoki \(2019\)](#), and the World Bank (2023). Bonds are discussed by [Heines and Jung \(2021\)](#) and [Schletz et al. \(2020\)](#), while art is noted by [Heines and Jung \(2021\)](#) and [Schletz et al. \(2020\)](#). [Morrow and Zarrebini \(2019\)](#) illustrate that tokenization can transform individual activities into digital tokens, as seen in projects like PlasticBank, which rewards recycling efforts; Power Ledger, which issues carbon credits for electric vehicle use; and Goodr, which tracks food donations to provide incentives or tax credits. Even personal healthcare data can be tokenized, as [Pithadia \(2023\)](#) explored, allowing varied ways to use and manage the data.

1.3. Blockchain adoption in the Philippines

The literature highlights the Philippines' significant progress in blockchain adoption. Compared to its counterparts in the Association of Southeast Asian Nations (ASEAN), such as Singapore, Malaysia, and Indonesia, the Philippines has emerged as a pioneer, particularly in the distribution of government bonds ([Ungson & Soorapanth, 2022](#)). [Ungson and Soorapanth \(2022\)](#) further note several local initiatives, including Project CyperOdin, which employs blockchain-based IoT (Internet of Things) to address environmental challenges in the Pasig River. Their study also cites efforts to position Mindanao and Northern Luzon as fintech hubs, alongside collaborations such as OneConnect and UBX Philippines Corporation's development of the country's first blockchain-enabled platform.

The World Bank (2023) also notes the Philippines' adoption of a regulatory sandbox, which offers a flexible framework to foster innovation in partnership with regulators. Globally, however, regulators have struggled to keep pace with the rapid evolution of blockchain technologies ([Ferreira, 2022](#); [The World Bank Group, 2023](#)). In the Philippines, the Bangko Sentral ng Pilipinas (BSP) oversees fintech and virtual currency exchanges through anti-money laundering regulations, while the Securities and Exchange Commission (SEC) licenses crypto trading platforms and provides guidance on initial coin offerings (ICOs) ([Lising, 2024](#)). Despite these advancements, [Lising \(2024\)](#), comparing U.S. and Philippine policies, observes that investors still favor the United States for tokenized infrastructure due to its perceived safety, underscoring the need to strengthen risk protections while continuing to promote innovation. As the Philippine SEC develops new rules for crypto-related services ([Mislos, 2025](#)), balancing these priorities becomes even more critical.

1.4. Examining the potential social risks of tokenized infrastructure

1.4.1. Transparency and immutability

Despite the salient features of blockchain, such as transparency, in which all transactions are visible to users ([Lapointe & Fishbane, 2018](#); [Schletz et al., 2020](#); [Uzsoki, 2019](#)), and immutability, which ensures that data cannot be altered or deleted once encoded on-chain ([Lapointe & Fishbane, 2018](#); [Pithadia et al., 2023](#); [Tian et al., 2022](#); [Tillemann et al., 2019](#)), researchers have pointed out the potential negative consequences of these features. [Lapointe and Fishbane \(2018\)](#) noted that transparency could expose personal information, risking exploitation

and persecution for vulnerable individuals, while immutability could lead to a “zero state problem (p. 18)” that happens when the accuracy of the information before entering the blockchain is questionable. While [Aljumah \(2023\)](#) notes that immutability limits individual influence that enhances business transaction control, [Viano et al. \(2023\)](#) argue that blockchain’s immutability can counteract the unpredictable nature of human relationships. The World Bank (2023) points out that the immutability feature increases the system’s vulnerability to cyber-attacks, exposing investors to fraud. [Schletz et al. \(2020\)](#) highlight that data quality on the blockchain hinges on accurate entry, directly impacting the intended purpose of transparency and immutability.

1.5. Decentralization and private keys

Decentralization, defined by [Holfmann et al. \(2020\)](#) as allowing for “transactions without [an] apparent central authority” (p.1), is another prominent feature of blockchain discussed by many researchers ([Baptista et al., 2023](#); [Lapointe & Fishbane, 2018](#); [Tian, Wang, et al., 2022](#)). However, [Viano et al. \(2023\)](#) argue that user-generated tokens do not guarantee full decentralized governance, noting that “some roles can still be centralized” (p. 9), which implies control by the tokenization platform makers. In addition to Viano et al., [Chung et al. \(2023\)](#) noted that tokenizing infrastructure, which involves linking off-chain data to the blockchain, can still be compromised by centralized control, undermining decentralization. Meanwhile, private keys, which are cryptographic codes used to access blockchain assets and enable decentralization, could also have negative consequences, as noted by researchers. [Lapointe and Fishbane \(2018\)](#) warn that losing a private key could permanently block access to property rights in blockchain systems, suggesting a need for secure access to private keys. [Tillemann et al. \(2019\)](#) raised a similar concern, indicating that if individuals were to obtain other people’s private keys, they might impersonate them and commit fraud.

1.5.1. Social inequality in tokenization

Building on the broader social implications of blockchain use, scholars have suggested that tokenization may disproportionately benefit wealthy individuals and institutions. [Lobban et al. \(2023\)](#), viewing the technology through the lens of an investment manager, argue that it “will not lead to a complete democratization of the asset class” (p. 19), instead primarily benefiting high-net-worth individuals. They further suggest that tokenization could “increase industry revenues by \$400 billion annually” (p. 18). Similarly, [Uzsoki \(2019\)](#) observes that tokenization has only been available to qualified investors, limiting access “to high-net-worth individuals and institutional investors” (p. 26). This disadvantage can be traced back to the limited access to education in economically disadvantaged regions. [Baptista et al. \(2023\)](#), [Domenicale et al. \(2024\)](#), and [Schletz et al. \(2020\)](#) identify the limited educational resources in poor areas as a key obstacle to the adoption of the technology. [Schletz et al. \(2020\)](#) note a shortage of skilled personnel hindering technology use in rural regions, while [Domenicale et al. \(2024\)](#) state that the technology requires a “high level of financial and digital literacy” (p. 17). Stressing the need for careful design of blockchain-based initiatives, [Lapointe and Fishbane \(2018\)](#) warn that the technology could “exacerbate existing negative social dynamics” (p. 17) if not implemented carefully, while [Tian, Wang, et al. \(2022\)](#) highlight that if a token acquisition is primarily capital-based, it could “assign more rights to the rich,” (p.17) further increasing inequality.

1.6. Distinguishing technological properties from design risks

Although often attributed to blockchain technology as a whole, the risks outlined in this paper require distinguishing between those intrinsic to its architecture and those introduced through specific design and implementation choices, as summarized in [Table 1](#). For example, transparency and immutability are core features of blockchain, but they

Table 1
Origins of risk in blockchain-based tokenized systems.

Risk theme	Origin of risk
Transparency	Inherent tech property: Risk arises when data privacy is poorly implemented on the blockchain ledger.
Immutability	Inherent tech property: Risk arises when inaccurate data is recorded without proper validation.
Private keys	Inherent tech property: Risk occurs if users lose access or face theft due to poor key management or interface design.
Tokenomics	Design risk: Risk arises when platform developers implement unsustainable reward models.
Decentralization	Design risk: Risk arises when platforms retain centralized control despite claims of distributed architecture.
Social inequality	Contextual design risk: Risk arises when system design results in disproportionate benefits for users.
Literacy gap	Contextual design risk: Risk arises when users lack access to onboarding, education, or digital literacy tools.
Regulatory uncertainty	Contextual design risk: Risk arises when legal frameworks fail to provide clear protections or oversight mechanisms.

Note. Inherent tech property refers to core features embedded in blockchain protocols, while contextual design risk refers to the interaction between blockchain systems and broader social, financial, or policy environments.

can produce harmful outcomes when platforms fail to incorporate privacy protections or verify off-chain data before encoding. Similarly, private keys are crucial for cryptographic security. However, issues such as lost access or fraud often arise not from the technology itself but from inadequate system design, including poor onboarding, insufficient recovery mechanisms, or confusing user interfaces. Additionally, while decentralization is often presented as a defining feature of blockchain, many platforms retain centralized control over token governance or infrastructure data linkage, undermining this principle in practice. Broader social outcomes, such as unequal access to tokenized systems or the exclusion of low-literacy populations, also reflect socio-technical design decisions rather than technical inevitabilities.

In addition to the social concerns raised by scholars, the lack of regulatory clarity worldwide regarding blockchain, driven by the technology’s novelty and rapid evolution, poses a significant obstacle to the broader adoption of tokenized infrastructure ([Uzsoki, 2019](#); [The World Bank Group, 2023](#); [Tian, 2020](#)). Research by [Nelson \(2019\)](#) saw that “more than 190 sovereign states” (p. 4) grapple with “a complex patchwork of government regulations” (p. 4), while [Lehmann \(2021\)](#) observed that countries struggle to establish firm plans for managing this novel technology. [Tian et al. \(2020\)](#) highlighted that an unclear regulatory climate may create arbitrage opportunities, leading to an unstable investment environment.

1.7. Blockchain implications in the Philippines

1.7.1. Digital competitiveness

The Philippines ranks 61st out of 67 countries globally and 13th out of 14 in the Asia-Pacific region in the IMD World Digital Competitiveness Ranking 2024, highlighting its lag in adopting advanced technologies. This limited digital competitiveness restricts the country’s potential for adopting technologies such as blockchain and tokenizing infrastructure, especially compared to its regional neighbors. The ranking evaluates the 63 economies’ ability to adapt and innovate with digital technologies, focusing on knowledge, technology infrastructure, and future-readiness metrics to highlight the strengths, gaps, and opportunities for digital growth.

The country’s overall low ranking can be traced to weaknesses across all three metrics. In the Technology category, the country ranks 56th overall, with particularly low scores in regulatory frameworks (66th), starting a business (65th), and enforcing contracts (64th), as well as intellectual property rights (59th). In the Knowledge category, the Philippines ranks 64th overall, with challenges in training and education (62nd) as evidenced by low rankings in public spending on education

(55th), higher education achievement (58th), and the percentage of women with tertiary degrees (60th). Lastly, in the Future Readiness indicator, the Philippines ranks 58th overall, with low scores in IT integration (58th), e-government (58th), government cybersecurity capacity (56th), and software piracy (56th). Low rankings in the Knowledge metric for the Philippines echo the challenges identified by Baptista et al. (2023), Domenicale et al. (2024), and Schletz et al. (2020), who highlight how limited educational resources and digital literacy in poorer areas create barriers to adopting technologies such as infrastructure tokenization.

1.7.2. Lower participation by women in education and crypto adoption

The Philippines' poor ranking for women with tertiary degrees and the significant gender gap in cryptocurrency ownership will exacerbate the existing gender inequalities introduced via the tokenization of infrastructure. The country ranks 60th in the IMD's digital competitiveness ranking for women with tertiary degrees, suggesting that tokenizing infrastructure could exacerbate gender disparities by limiting access for women without degrees in higher education. A similar finding appears in a study by Finder (2023), an Australian fintech research company, which shows a significant gender gap in crypto ownership, with 33 % of Filipino men owning cryptocurrency compared to 22 % of women. In contrast, South Korea demonstrates a narrower gap, with 24 % of men and 23 % of women holding crypto.

1.7.3. Financial literacy

The low knowledge score of the Philippines in the IMD's competitiveness ranking is reflected in the below-average rating of financial literacy, as ranked by the OECD/INFE 2023 *International Survey of Adult Financial Literacy*, a policy paper that assesses financial knowledge, behaviors, and attitudes across countries. The Philippines scores 58 out of 100 in financial literacy, below the global average of 60 and the OECD average of 63. In comparison, Germany (76), Hong Kong (70), and Finland (67) have higher scores, while Jordan (58), Indonesia (57), and Paraguay (50) score similarly or lower, with Yemen holding the lowest score of 42. Such low financial literacy, which is related to broader educational limitations (Cancio & Romarate, 2024), illustrates the concerns of Lapointe and Fishbane (2018), Domenicale et al. (2024), and Baptista et al. (2023) that limited financial and digital literacy can hinder safe access to, understanding of, and participation in blockchain systems. A striking example of the Philippines' limited financial literacy comes from the same OECD database, which shows that only 25–30 % of Filipino crypto-asset holders correctly understand that cryptocurrencies are not legal tender.

1.7.4. Access to technology

Concerns about the tokenization of infrastructure potentially benefiting the wealthy and reinforcing economic disparities, as raised by Lobban et al. (2023), Uzsoi (2019), and Tian et al. (2022), are reflected in the Philippines' broadband affordability data. The country ranks 104th in Internet Affordability in the 2023 Surfshark Digital Quality of Life Index (Surfshark, 2023), down from 72nd in 2021 and 98th in 2022. This decline mirrors broadband affordability issues highlighted in the 2022 Global Connectivity Report by the International Telecommunication Union (ITU), which notes that mobile broadband prices exceed the 2 % affordability target for those occupying the lower 40 % income group, hindering equitable access.

1.7.5. Potential for leapfrogging technology

While previous analysis highlights the Philippines' vulnerability to social challenges, other datasets suggest the country has the potential to leapfrog by successfully implementing infrastructure tokenization. Orosa-Ople (2023), citing the Bangko Sentral ng Pilipinas (BSP) data, reports that e-wallet accounts in the Philippines grew from 5.7 million in 2019 to 258 million by 2022, and by then, the number of e-wallet users had surpassed bank account holders, with 27.5 million e-wallet users

compared to 18 million bank account holders. These data suggest that many Filipinos have bypassed traditional financial systems in favor of digital finance and that they have become more familiar with e-wallets and mobile payments, thus laying the groundwork for greater adoption of cryptocurrencies and tokenized systems.

Following this trajectory, other emerging trends suggest that the Philippines, through its progress in crypto adoption, tech research, and infrastructure development, is poised to lead in crypto adoption, as highlighted by blockchain analytics firm Chainalysis in its 2024 Global Crypto Adoption Index. The Philippines ranks eighth globally, with approximately \$40 billion in crypto value received from July 2023 to June 2024, placing it behind other Central and Southern Asia and Oceania (CSAO) countries such as Indonesia, India, and Vietnam. Moreover, the 2024 IMD digital competitiveness ranking highlights the Philippines' strong representation of female researchers, ranked second globally, suggesting that despite a lower ranking for women with tertiary degrees, the high representation of women in tech research could help address gender gaps and encourage greater female participation in infrastructure tokenization. The Philippines' strengths in crypto adoption and female representation in tech research are supported by progress in infrastructure development. The IMD digital competitiveness ranking places the Philippines ninth in Investment in Telecommunications, aligning with World Bank (2024) data that ranks the country sixth in mobile phone subscriptions per 100 people in the Asia-Pacific.

Adding to these positive metrics, Kapron (2023) highlights the Philippines as the most pro-crypto country in Southeast Asia, supported by regulatory tolerance tied to BSP's digitization and financial inclusion goals. Studies by Baker McKenzie (2020) and the World Bank (2023) further note that the BSP's sandbox fosters blockchain innovation as the country drafts clearer policies. Moreover, the country ranks 31st in Public-Private Partnerships (PPP) and 22nd in Attitudes Toward Globalization, indicating a favorable environment for collaboration and global integration.

1.8. Axie infinity and social implications of tokenization

The failure of Axie Infinity, a blockchain-based play-to-earn game in which players earn cryptocurrency convertible to cash, offers a real-world example of how a blockchain application has exacerbated social vulnerabilities in the Philippines. At its peak, Axie Infinity attracted over 2.7 million daily active users (Time, 2022), particularly drawing in players from developing countries such as the Philippines, Brazil, and Venezuela, who had lost their jobs due to the impact of the COVID-19 pandemic in 2020 (Fortune, 2021). Users in the Philippines once constituted 40 % of the global players (Time, 2022). Fortune (2021) reported that Axie facilitated about \$30 million in daily token transfers, an amount small in comparison to the \$2.2 trillion crypto market in 2021 but, as the magazine observed, "meaningful for players—and governments—in poorer countries."

To play the game, users must purchase digital creatures called Axie monsters or borrow them through the system's manager-scholar model, where "managers" (existing players) lend Axies to new players or "scholars." Scholars use the borrowed Axies to participate in battles, earning Smooth Love Potion (SLP) tokens as rewards. These earnings are then split between the manager, who provides the Axie monsters, and the scholar, who plays the game. According to a profile by Reuters (2023), Axie Infinity managers in the Philippines included individuals from high-earning professions, such as doctors and businessmen. Time (2022) reported that one such manager was based in Australia, while his scholar was located in the Philippines, indicating that individuals from wealthier nations can tap into the resources of developing countries. In contrast, Reuters (2023) noted that scholars were typically high school students and gig workers from poorer provinces. One scholar earned \$487 in 15 days and spent \$1200 to purchase his own Axie monster (Time, 2022). This contrasts with a manager who earned \$10,000 per day for having 200 Axie scholars (Reuters, 2023). Power dynamics

exacerbate this social gap, as one manager required a scholar to spend approximately five to six hours of gameplay while taking half of the earnings (Time, 2022), and another took a 60 % cut of players’ income (Reuters, 2023).

The social implications of Axie Infinity became more evident when the system crashed in 2022 following a major security breach in March and a broad cryptocurrency market decline, compounded by issues with its tokenomics or how its token system was designed to work. As Time (2022) noted, the game’s reliance on a continual influx of new players to sustain the value of SLP is a model that developers acknowledged in the white paper, comparing such tokenomics to a Ponzi scheme. This reliance on new players supports Viano et al. (2023), who argue that user-generated tokens do not guarantee decentralization, as control over the SLP token’s value remains with the developers. Media reports indicated that both managers and scholars lost money after the price of the SLP token dropped from \$0.36 to about \$0.003, a dramatic loss in value of approximately 99.2 %. However, it can be inferred that scholars were more affected, as the game was their primary source of income. KrAsia (2022) noted that play-to-earn games are key entry points into crypto for those with limited financial literacy, indicating players were unaware of the risks, as many lost everything they had earned and even went into debt (Reuters, 2023; Time, 2022). Lastly, a review of media articles on the Axie Infinity case study yields no evidence supporting Lapointe and Fishbane’s (2018) concerns about privacy exposure or Tillemann et al.’s (2019) issues with private key theft and fraud.

The Axie Infinity case illustrates how social risks linked to tokenized

infrastructure manifest in practice, as summarized in Table 2. While issues such as privacy exposure and private key theft were not prominently reported in this context, significant risks arose from the system’s token design, governance structure, and user engagement model. In particular, Axie’s economic framework intensified social inequality, its governance concentrated control over token value in the hands of developers, and its reliance on continuous player growth demonstrated speculative dynamics. These traits closely align with concerns regarding centralization, exploitative tokenomics, and the exclusion of users with limited financial literacy, underscoring how ill-conceived tokenized systems can exacerbate existing vulnerabilities rather than enhance access.

2. Discussion

While not yet fully operational worldwide, the tokenization of infrastructure represents a promising option for addressing the significant funding gaps in infrastructure projects, particularly in developing countries. With blockchain applications already advancing in sectors such as real estate, art, and finance (Heines & Jung, 2021; Tian et al., 2020), the potential for tokenizing infrastructure is increasingly within reach. However, as blockchain systems evolve, scholars have warned that they may exacerbate existing social inequalities, particularly for marginalized groups, if blockchain initiatives are not carefully designed (Lapointe & Fishbane, 2018; Tian et al., 2022). Tokenization of infrastructure, as an emerging application, may follow the same trajectory, with disproportionate impacts on vulnerable populations. This paper demonstrates how some of these concerns are already evident in earlier blockchain applications and argues that they will be further magnified with the widespread adoption of infrastructure tokenization in developing economies such as the Philippines.

The meta-analysis of the Philippines’ digital landscape provides evidence of existing challenges in digital and financial literacy and infrastructure readiness, findings that align with experts’ concerns that these weak spots may heighten the social risks tied to blockchain technologies, including infrastructure tokenization. For example, the country’s low rankings in enforcing contracts and intellectual property rights, software piracy, and government cybersecurity capacity reflect the concerns of Uzsoki (2019), Lapointe and Fishbane (2018), and the World Bank Group (2023) on the need for robust legal protections to build trust and protect consumers in tokenized systems. Similarly, the case of Axie Infinity provides further evidence of how the tokenization of infrastructure could unfold, particularly in developing economies. In the case of Axie, many low-income individuals were drawn to the opportunity for quick financial gains without fully understanding the risks involved, ultimately leading to significant losses (Fortune, 2021; Reuters, 2023). This case illustrates how tokenization offers the potential for economic inclusion but also risks deepening social inequalities, as wealthier individuals and institutions may exploit vulnerable populations for profit, as noted by Tian, Wang, et al. (2022) and Lapointe and Fishbane (2018). While Axie Infinity does not offer fractional ownership, it shares similarities with infrastructure tokenization in its use of digital assets and a decentralized system in which participants can trade and earn tokenized assets. Both systems expose vulnerable participants, those with lower financial literacy or weaker digital skills, to exploitation by wealthier, more knowledgeable actors, potentially widening social inequalities.

3. Conclusion

This paper’s meta-analysis and proxy case study address the gap in empirical evidence in the literature on blockchain’s potential and its social impact, particularly in relation to the widening of income inequalities in emerging economies. It sheds light on how blockchain-based tokenization interacts with existing inequalities in regions like the Philippines, offering a nuanced understanding of these impacts, particularly in the context of the Philippines’ unique economic and

Table 2
Summary of risks and recommendations from the axie infinity case study.

Risk Theme	Evidence from the Axie Infinity Case Study	Prescriptive Insights
Transparency	No reports were found of privacy violations or harmful data exposure.	No incidents observed; maintain privacy safeguards and continue monitoring.
Immutability	No mention of harmful permanent data or input errors in the case study.	No risks surfaced; uphold existing immutability controls and consider correction options.
Private Keys	No cases of key loss or theft were reported.	No incidents reported; promote best practices in key management and user education.
Tokenomics	Reports noted a reliance on continual new user growth to sustain SLP value, likening the tokenomics model to a Ponzi scheme.	Tokenomics should ensure fairness and long-term sustainability.
Decentralization	Reports indicate that developers retained control over tokenomics and reward mechanisms, including decisions regarding token value and allocation.	Governance should be more transparent to improve fairness.
Social Inequality	Reports show earnings disparities, with users in lower-income regions and non-managerial roles earning less.	Future projects should examine designs that amplify or reduce social inequality.
Literacy Gap	Reports indicate that users lack awareness of associated risks.	Project providers or governments must ensure that users possess basic financial literacy and understand the associated risks before participating.
Regulatory Uncertainty	Regulatory responses were not observed or explored within the scope of this case study.	Regulatory issues were not deeply explored; future research should address this gap.

Note. This table summarizes key risk themes from the Axie Infinity case study, with evidence and prescriptive insights. For risks without reported incidents, recommendations focus on monitoring. Regulatory uncertainty was not deeply explored and warrants future study.

social challenges. The findings, particularly from the Axie Infinity case study, provide evidence that poorly designed blockchain-based systems can reinforce existing social inequalities, a concern raised by Lapointe and Fishbane (2018) and Tian, Wang, et al. (2022). While focused on a single case to provide a detailed analysis, these findings contribute to foundational knowledge that can inform broader inquiries. A key direction for future study is to examine governmental or regulatory responses to similar events, as well as to analyze on-chain data specific to the Philippines that were not covered due to feasibility constraints. Lastly, the framework on risk themes and origins offers a valuable tool for assessing tokenization projects.

Ethics in publishing statement

This research presents an accurate account of the work performed, all data presented are accurate and methodologies detailed enough to permit others to replicate the work.

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Data availability

All data analyzed in this study are publicly available and have been cited in the References section. No additional datasets were generated for this study.

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